

DEEPAK YADAV

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EDUCATION

B.E. in Electrical & Electronics | Delhi Technological University (formerly DCE)

May'13-Jun'17

Coursework: Electronic Circuit Analysis & Design, Digital System Design, Analog Electronics, Digital Electronics, Communication systems, Power Electronics, Signal & Systems, Microcontrollers, Embedded systems.

TECHNICAL SKILLS

- **Languages/Protocols/Software Tools:** C/C++, PCIe (NVMe), SD/uSD, Visual Studio Code, Visual Studio 2017/19.
- **Core Competencies:** System Design & Technical Communication, NVMe/PCIe Systems & Performance, Digital Signal Processing, Communication Systems, NAND Flash Fundamentals, SSD Architecture & Data Path Design.
- **Hardware Tools/Analog Designs:** JTAG, Intel VTune, Go-logic, Prodigy (PGY-SSM), LeCroy (T48/T54), Viavi (Xgig5P8/5P16), Quarch (QPS), Digital Oscilloscopes, AWGs, Digitizers, Spectrum Analyzers.

PROFESSIONAL EXPERIENCE

Micron | Lead, Systems Software Engineer

Sep'2025-Present

- **Led system-level architecture, host, power, performance, thermal, and competitive analysis** for SSD platforms, driving data-driven optimizations across hardware and software stacks.
- **Analyzed competitor SSD behavior under application-driven and AI training workloads** using industry benchmarks to characterize latency, throughput, QoS, and sustained performance.
- **Performed system and host-side performance characterization** using Linux and Windows environments, correlating CPU utilization, PCIe/NVMe stacks (I/O submission/completion paths), memory behavior, and device interactions to identify performance and efficiency bottlenecks.
- **Enabled AI training and inference workflows by characterizing** SSD behavior in data-loading pipelines, checkpointing, and mixed read/write scenarios common to modern ML frameworks.
- **Domains:** System & Host Analysis (OS, Driver, CPU, Memory), SSD & NVMe Architecture, PCIe Performance & Power States, AI Training & Inference Storage Workloads, Competitive Benchmarking & Characterization, System Workload Modeling, Sustained Performance & QoS Analysis.

Western Digital | Staff, Systems Design Engineer

July'2022– Dec'2024

- **Key contributor to the Removable Products Group (RPG)**, System Design team (SD/uSD), driving system-level **feature design** and development to meet increasing capacity and performance demands.
- **Authored detailed system and feature specifications**, collaborating cross-functionally with firmware, ASIC design, validation, and NAND technology teams to ensure end-to-end alignment.
- **Successfully delivered a 1.5TB uSD card**—the highest capacity in the smallest form factor—by resolving complex challenges related to thermals, system performance, reliability, and host compatibility.
- **Improved uSD read/write throughput** by optimizing firmware data paths and reordering NAND access sequences, resulting in higher sustained performance and improved efficiency. Reviewed competitor SD/uSD products to benchmark capacity, performance, and feature differentiation, influencing roadmap and feature prioritization.
- **Analyzed sustained performance degradation mechanisms** (thermal throttling, write amplification, EPWR, Wear leveling, garbage collection, DPDM) and proposed firmware and system-level mitigations.
- **Led host compatibility and interoperability analysis** across diverse platforms (mobile, camera, embedded, PC), ensuring standards compliance and robust real-world behavior.
- **Domains:** SD/uSD System Architecture, Block Budget & Performance estimation, Feature algorithms, Firmware Data Path, Power-Performance-Thermal Trade-offs, Sustained Performance & Reliability, Host Compatibility, Validation & Qualification.

Keysight Technologies | RnD Engineer II

Apr'2018– July'2022

- **Owned end-to-end system integration** for high-performance AWG and VSG platforms (M8190A, M8195A, M8196A, M8121A, M9383, M8131A), including firmware, SFP interfaces, SCPI command APIs, and hardware drivers.
- **Partnered with global, cross-functional engineering teams** in the US and Germany to validate signal integrity, ensure platform stability, and maintain measurement accuracy across instrument platforms.
- **Performed signal integrity (SI) analysis and validation** for high-speed DAC/ADC paths, clocks, probes, and I/O interfaces, ensuring waveform fidelity, low jitter, and measurement accuracy.
- **Supported New Product Introduction (NPI) execution and system-level validation activities**, including integration testing, defect resolution, and enabling delivery of high-quality, production-ready test and measurement instruments.
- **Domains:** Test & Measurement Systems Architecture, Firmware Validation, Instrument Control (SCPI), Signal Integrity, Hardware-Software Integration, NPI Execution, DAC/ADC, DSP & Clocking Architectures, Probes & Measurement Accuracy, Firmware & FPGA Validation, Jitter & Phase Noise Analysis.

PATENT APPLICATIONS

- **US Patent Application 2022-INV-0001665** – *Flash Memory Read Scrub with Minimized Read Amplification* (Filed Nov 03, 2022)
Inventors: Deepak Yadav, Ajay Singh
- **US Patent Application 2023-INV-0001850** – *Stream Detection with Minimized Read Amplification* (Filed Feb 10, 2023)
Inventor: Deepak Yadav